An Efficient Object Detection Model Based on NanoDet

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***Abstract*—Object detection is a fundamental task in computer vision, involving the prediction of bounding boxes and class labels for Regions of Interest (ROI) within images. Traditionally, anchor-based detectors have dominated this field, leveraging predefined anchor boxes for object localization and classification. However, anchor-free detectors have emerged as a compelling alternative due to their reduced computational complexity and efficient detection capabilities. Among these innovations, NanoDet has garnered significant attention in the AI community for its speed and lightweight design as an anchor-free object detection model. This study introduces an enhanced variant of NanoDet, leveraging the EfficientNetB0-compact model as its backbone. By replacing the original ShuffleNetV2 backbone, this integration aims to improve detection performance without compromising speed, thereby enhancing suitability for mobile devices. Experimental results demonstrate the effectiveness of this modification in achieving superior performance metrics, highlighting the potential of advanced network architectures in optimizing object detection tasks for mobile and edge computing environments.**

***Index Terms*—computer vision, object detection, efficient, anchor-free, EfficientNetB0-compact, nanodet**

I. Introduction

Object detection is pivotal for applications like autonomous driving and surveillance, aiming to localize and classify objects in images or video frames. Anchor-based detectors such as Faster R-CNN [12], YOLO [11], and SSD [8] have traditionally dominated the field, using predefined anchor boxes to propose object regions for refinement and classification.

Despite their effectiveness, anchor-based detectors are computationally demanding, especially on mobile and embedded systems. Anchor-free methods like FCOS [16], YOLOv7 [18], YOLOv5 [5], and YOLOv8 [4] address these challenges by eliminating anchor boxes, predicting object bounding boxes and classifications directly, thus reducing computational complexity [1].

NanoDet [10], designed for mobile platforms, leverages the FCOS framework and incorporates techniques like the Assign Guidance Module (AGM) and Dynamic Soft Label Assigner (DSLA) to enhance label assignment and detection precision, achieving 97 FPS on mobile ARM CPUs with a lightweight model footprint. EfficientNet [14], developed through AutoML, combines MobileNetV2's inverted bottleneck residual blocks with squeeze-and-excite mechanisms for a balance of accuracy and computational efficiency across scales (EfficientNetB0 to EfficientNetB7) [3]. EfficientNetB0 is prominent for its compact size and high performance on mobile devices [15], using compound scaling to adjust network depth, width, and resolution.

In this study, we enhance NanoDet by replacing its backbone, ShuffleNetV2 [9], with EfficientNetB0-compact [15], [17]. Our experiments show significant improvements in detection accuracy without sacrificing speed, demonstrating the benefits of combining lightweight, efficient models with robust detection frameworks for superior performance in mobile and edge computing environments.

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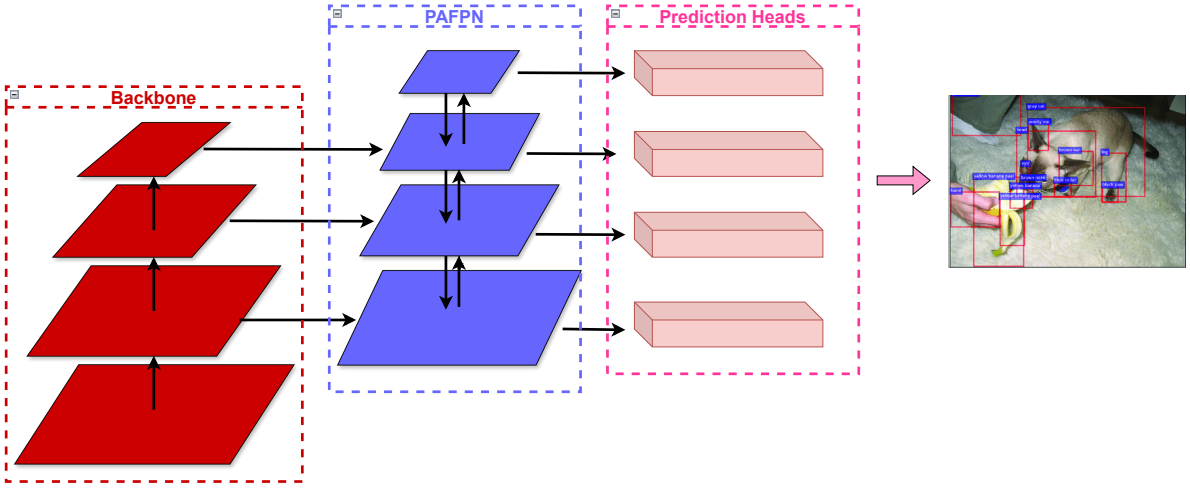


Fig. 1. Architecture of the baseline Nanodet [10]. It consists of a backbone, a path aggregation feature pyramid network (PAFPN) and several prediction heads.

II. Related Work

A. Efficient object detectors

NanoDet [10] is a highly efficient, lightweight, anchor-free object detection model designed specifically for mobile devices. Unlike traditional anchor-based models, NanoDet eliminates the need for predefined anchor boxes, significantly reducing computational complexity and improving detection efficiency. The model is built on the FCOS framework and includes enhancements such as the Assign Guidance Module (AGM) and Dynamic Soft Label Assigner (DSLA) to refine label assignment and boost detection accuracy.

The upgraded version, NanoDet-Plus [10], achieves real-time performance on mobile ARM CPUs, running at an impressive 97 FPS. This makes it an ideal solution for resource-constrained environments, requiring minimal GPU memory for training and supporting multiple deployment backends like ncnn, MNN, and OpenVINO.

One of the standout aspects of NanoDet is its ability to maintain high accuracy despite its compact size—980KB (INT8) or 1.8MB (FP16)—making it suitable for deployment on low-cost edge devices. The model performs exceptionally well on benchmark datasets like COCO, demonstrating robust detection capabilities.

PP-PicoDet [19] is a lightweight object detection model by Baidu's PaddlePaddle team, optimized for mobile and edge devices. It balances speed, accuracy, and size using key techniques.

The model features a Path Aggregation Network (PAN) for multi-scale feature extraction and efficient backbones like ShuffleNetV2 and MobileNetV3. Advanced loss functions like Generalized Intersection over Union (GIoU) and focal loss enhance accuracy, while Non-Maximum Suppression (NMS) refines detection results.

PP-PicoDet supports quantization to reduce model size and inference latency. It achieves competitive accuracy with lower computational costs, making it ideal for real-time applications on resource-constrained devices.

B. Efficient Convolution module

ShuffleNet [20] and its successor ShuffleNetV2 [9] are pioneering convolutional neural networks explicitly crafted to excel in efficiency and performance on mobile and embedded devices. The original ShuffleNet innovates by integrating pointwise group convolution and channel shuffling techniques, which significantly decrease computational costs while preserving high accuracy. This approach divides input channels into groups and shuffles them, ensuring effective information exchange across these groups. Building upon these foundations, ShuffleNetV2 further optimizes memory access and device-specific performance considerations. It retains the fundamental innovations of its predecessor while introducing additional advancements such as channel splitting, which enhances parallelism and reduces overhead even further. These combined features position both ShuffleNet and ShuffleNetV2 as optimal choices for real-time applications operating on hardware with constrained resources.

Depthwise separable convolutions, as introduced in MobileNetV1 [2], represent a breakthrough in efficient convolutional operations. They partition spatial filtering and feature generation into separate layers: a lightweight depthwise convolution captures spatial information efficiently, followed by a pointwise convolution that aggregates cross-channel information seamlessly.

MobileNetV2 [13] introduces the novel concept of a linear bottleneck and inverted residual structure, designed to enhance efficiency by leveraging the low-rank nature of convolutional operations. This architecture incorporates an expansion convolution, depthwise convolutions, and a projection layer, with residual connections used when input and output dimensions match. Such design choices ensure compact representations while enabling powerful nonlinear transformations internally.

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EfficientNets [14], a family of convolutional networks optimized for image classification, have emerged as state-of-the-art models due to their meticulous balance of depth, width, and resolution through compound scaling. This approach effectively reduces model size and computational requirements while preserving high performance.

At the heart of EfficientNets lies the inverted bottleneck convolution module, inspired by MobileNetV2's linear bottleneck and inverted residual structure. This module efficiently captures and propagates feature maps across various scales and resolutions, contributing significantly to the overall efficiency and effectiveness of the model.

III. Proposed Method

The architecture of NanoDet is meticulously crafted to prioritize lightweight efficiency, rendering it exceptionally suitable for deployment on mobile devices. Its current backbone, ShuffleNetV2 [9], was chosen for its robustness and cost-effectiveness within resource-constrained environments. The neck of NanoDet employs a simplified Path Aggregation Feature Pyramid Network (PAFPN) [7], which optimizes by utilizing only 1×1 convolutions, ensuring a leaner model structure. In the head of NanoDet, depth-wise convolutions further enhance its lightweight design. This design choice enables simultaneous computation of border regression and classification tasks through shared convolutional layers, split into optimized components for computational efficiency. For detailed insights into NanoDet's architecture, refer to Figure 1.

EfficientNet-B0 [14], the foundational model in the EfficientNets series, incorporates the mobile inverted bottleneck convolution module and integrates an attention mechanism via compression and excitation modules. Recognizing the demand for models optimized for mobile and IoT applications, the authors introduced an EfficientNet-B0 lite version by omitting the squeeze-and-excite (SE) module [3] and substituting swish activations with ReLU6 to facilitate post-quantization.

Building upon this groundwork, Thanh et al. [17] refined the model further to create EfficientNet-B0-Compact. This variant implements specific optimizations aimed at enhancing efficiency and compatibility. Notably, it removes the initial MBConv1 block at stage 1, based on findings that early high-resolution operations significantly impact computational load without substantial accuracy loss. Additionally, the second MBConv6 block is replaced with a Conv31 block comprising a 3×3 convolution followed by a 1×1 convolution, further streamlining operations.

These refinements collectively contribute to an architecture that is not only more efficient but also highly suitable for mobile and IoT deployments. A detailed structural breakdown of EfficientNet-B0-Compact can be found in Figure 2.

In this study, we propose replacing the ShuffleNetV2 backbone in NanoDet with EfficientNet-B0-Compact. This modification aims to enhance overall performance while maintaining comparable accuracy levels.

IV. Experiments

A. Dataset

The proposed model is evaluated using the Microsoft Common Objects in Context (MS COCO) dataset [6], a widely recognized large-scale dataset extensively annotated for various computer vision tasks, including object detection, instance segmentation, and image captioning. Introduced in 2014, MS COCO has established itself as a pivotal benchmark for advancing state-of-the-art research in these fields.

For this study, we utilize the 2017 version of the MS COCO dataset, which comprises 118,000 images (train2017) for training and 5,000 images (val2017) for validation purposes. This dataset encompasses a diverse array of scenes and objects, annotated with bounding boxes for 80 different object categories. While MS COCO serves multiple computer vision tasks, our focus in this paper is specifically on object detection, leveraging its comprehensive annotations and benchmarks tailored for evaluating detection model performance.

B. Implementation Details

In this study, all models were implemented using the code base of NanoDet [10]. The training phase spanned 300 epochs, employing a batch size of 128 on a server equipped with an AMD Ryzen 9 3900XT CPU and a GeForce RTX 3090 GPU.

Furthermore, the inference speed of each model was evaluated using a batch size of 1 on the same hardware configuration. This standardized evaluation facilitated a fair comparison of computational efficiency and real-time performance across different models, offering valuable insights into their suitability for practical deployment.

Throughout the experiments, the input images were uniformly resized to dimensions of 320×320, ensuring consistent conditions for model evaluation and comparison.

C. Ablation Studies

Followed [17], we also do some ablation studies to check whether the trend in their results also work on our model. The ablation study explores the impact of modifying the backbone from EfficientNetB0 to a more compact version, shedding light on the trade-offs between performance and efficiency. Incorporating EfficientNetB0 significantly boosts detection performance, with $\text{mAP}^{50:95}$ improving from 25.64% to 31.25%, though at the cost of increased parameters and slower CPU inference. Removing the first block slightly reduces computational complexity while maintaining competitive performance, suggesting that the initial layers contribute modestly to overall accuracy. Meanwhile, replacing the early-stage layers with a Conv31 reduces FLOPS more noticeably, enhancing inference speed on both CPU and GPU while maintaining respectable accuracy. These findings highlight that strategic adjustments to the backbone can effectively balance accuracy and computational efficiency, offering practical pathways for optimizing lightweight models in resource-constrained environments. So it easy to see that the trend in [17] not only work with image classification but also object detection task.

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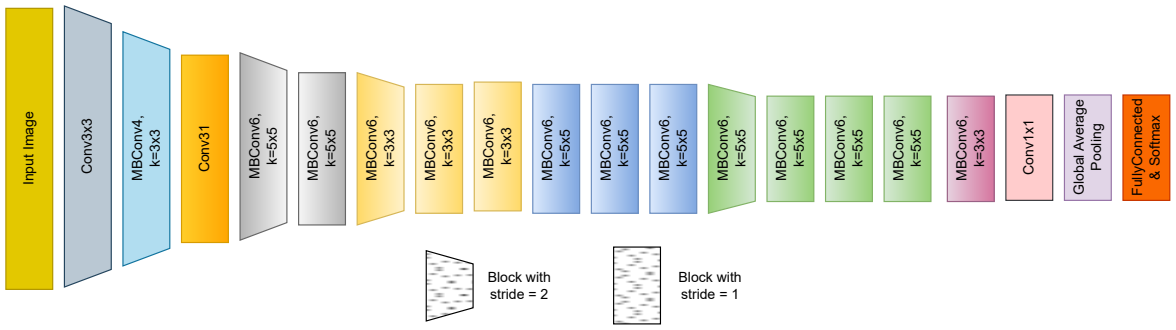


Fig. 2. Architecture of the EfficientNetB0-compact. **Conv31**: Convolution block with a 3×3 convolution layer and a 1×1 convolution layer. **MBConv**: Mobile inverted Bottleneck Convolution, the following number is the expanded factor.

TABLE I
ABLATION STUDIES ON MODIFYING THE BACKBONE FROM THE EFFICIENTNETB0 TO EFFICIENTNETB0-COMPACT.

Model	#Params (10^6)	#FLOPS (10^9)	mAP$^{50:95}$ (%)	mAP^{50} (%)	mAP^{75} (%)	mAP^{S} (%)	mAP^{M} (%)	mAP^{L} (%)	Speed-CPU (ms)	Speed-GPU (ms)
Baseline [10]	1.17	0.9	25.64	39.67	26.69	7.56	26.21	42.92	21.28	8.57
EfficientNetB0 as backbone	3.30	1.86	31.25	47.17	32.76	11.28	33.71	51.85	28.35	7.91
Remove first block	3.30	1.84	30.83	46.71	32.30	10.73	33.09	50.69	27.44	7.80
Use Conv31 in stage 1	3.29	1.68	29.51	45.02	30.79	8.90	30.88	49.73	22.53	7.52

D. Experiment Results

Table II presents a comprehensive comparison of the proposed model against NanoDet, PP-PicoDet, and other related models on the COCO object detection task. The results highlight the proposed model's ability to achieve a strong balance between accuracy and efficiency, outperforming several baselines in terms of Mean Average Precision (mAP) across different metrics, while maintaining competitive inference speeds on both CPU and GPU.

When comparing the proposed model to NanoDet-Plus-m, a widely-used lightweight detector, the improvements are significant across multiple metrics. The proposed model achieves a 2.9% higher mAP$^{50:95}$, a 5.5% higher mAP^{50}, and a 4.1% higher mAP^{75}, showcasing its ability to detect objects with greater precision across varying Intersection over Union (IoU) thresholds. Additionally, the proposed model shows improved performance across different object sizes, with noticeable gains particularly in detecting large objects — achieving a 6.81% improvement in mAP^{L}. This indicates that the model effectively captures high-level features, enhancing its robustness in scenarios where objects vary in scale.

Interestingly, when comparing the proposed model to NanoDet-Plus-m-1.5x, which has a similar computational cost, the proposed model achieves slightly lower mAP scores. However, it is crucial to consider that NanoDet-Plus-m-1.5x benefited from pretrained weights on the ImageNet dataset, while the proposed model was trained from scratch. Pretraining on large-scale datasets like ImageNet often provides a significant head start in feature extraction, especially for complex tasks like object detection. Given this, it is reasonable to expect that incorporating pretrained weights into the proposed model could further boost its performance, potentially closing the gap or even surpassing NanoDet-Plus-m-1.5x.

The results also emphasize the model's efficiency. Despite having a slightly higher parameter count and FLOPS than some NanoDet variants, the proposed model maintains competitive inference speeds. On the CPU, it achieves an inference time of 22.53 ms, only marginally slower than NanoDet-Plus-m, while being faster on the GPU at 7.52 ms. This balance of speed and accuracy makes the model well-suited for real-time applications, where both factors are crucial.

In summary, the proposed model demonstrates notable improvements in detection accuracy, especially for larger objects, while maintaining efficiency. Its superior performance across mAP metrics without leveraging pretrained weights highlights its capacity to learn effective feature representations from scratch. Future work could explore integrating pretrained weights to further push performance boundaries. These findings suggest that the proposed model holds significant promise for lightweight object detection in resource-constrained environments, providing a robust and efficient solution for real-world applications.

V. CONCLUSION

This paper presents a novel, efficient object detector by integrating the EfficientNet-B0-Compact as the backbone of NanoDet, replacing the original ShuffleNetV2. The architectural modifications introduced in the proposed model have significantly enhanced its representational capacity and object detection capabilities. As a result, the model achieves superior performance metrics while maintaining high speed, making it particularly suitable for mobile devices. Experimental results validate the effectiveness of this modification, underscoring the potential of advanced network architectures in optimizing

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TABLE II
EXPERIMENTS RESULTS OF NANODET AND PROPOSED MODEL ON OBJECT DETECTION TASK OF COCO DATASET.

Model	#Params (10^6)	#FLOPS (10^9)	mAP$^{50:95}$ (%)	mAP^{50} (%)	mAP^{75} (%)	mAP^{S} (%)	mAP^{M} (%)	mAP^{L} (%)	Speed-CPU (ms)	Speed-GPU (ms)
PP-PicoDet-S* [19]	0.99	0.73	27.1	41.4	-	-	-	-	-	-
NanoDet-EfficientLite-B0* [10]	3.11	1.72	24.7	-	-	-	-	-	-	-
NanoDet-Plus-m-1.5x* [10]	2.44	1.75	29.9	-	-	-	-	-	25.27	8.70
NanoDet-Plus-m* [10]	1.17	0.9	27.0	-	-	-	-	-	21.28	8.57
NanoDet-Plus-m** [10]	1.17	0.9	26.95	41.33	28.29	7.73	28.07	45.51	21.28	8.57
NanoDet-Plus-m [10]	1.17	0.9	25.64	39.67	26.69	7.56	26.21	42.92	21.28	8.57
Proposed model	3.29	1.68	29.51	45.02	30.79	8.90	30.88	49.73	22.53	7.52

* denotes model download from the author.
** denotes model trained with pretrained weight from external dataset.
Other models are trained from scratch, without pretrained weight.

object detection tasks for mobile and edge computing environments. This work highlights the synergy between lightweight, efficient models and robust detection frameworks, paving the way for more efficient and accurate real-world applications.

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